ASSIGNMENT NO. : 04

PRACTICAL: DevOps and CI/CD in Machine Learning

Aim

To study how DevOps principles, especially CI/CD, improve the deployment, updating, and maintenance of Machine Learning projects.

Introduction

DevOps combines development and operations to automate software development, testing, and deployment. In ML projects, these practices are called **MLOps**. They help manage code, data, models, testing, deployment, and monitoring.

DevOps Principles in ML

1. Version Control

Git/GitHub tracks changes in code, datasets, configurations, and model versions.

2. Continuous Integration (CI)

Automatically tests the ML code whenever new changes are pushed.

`Code → Build → Test → Validate`

3. Continuous Delivery/Deployment (CD)

Automatically packages and deploys the tested ML model to the required environment. `Tested Model → Package → Deploy → Production`

4. Continuous Training

Automatically retrains the model when new data becomes available or model performance decreases.

5. Monitoring

Monitors model performance, data changes, errors, and system performance after deployment.

Benefits

- Faster deployment
- Automated testing
- Easy model updates

- Fewer manual errors
- Better version control
- Continuous monitoring
- Improved reliability and reproducibility

Result

DevOps and CI/CD help automate the **testing, deployment, updating, and monitoring** of Machine Learning models.